AC9M6SP01 • Year 6 Mathematics

Parallel Cross-sections of Three-dimensional Objects

Predict and recognise how slices relate to an object's structure

We are learning to…

Students visualise and construct slices parallel to a base, identify when cross-sections remain congruent and compare prisms, cylinders, pyramids and cones.

Represent | Reason | Calculate | Interpret

Verify

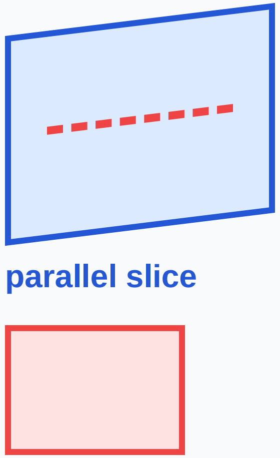
Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

recognise and describe the parallel cross-sections of objects and recognise their relationships to the object's base and shape

Slice a prism parallel to its base



Every parallel slice of a right rectangular prism is a congruent rectangle

For a prism or cylinder, cross-sections parallel to the base keep the base shape and size. For a pyramid or cone, parallel sections keep the shape but change size.

Compare constant and changing cross-sections

object	base	parallel cross-section
rectangular prism	**rectangle**	**congruent rectangle**
triangular prism	**triangle**	congruent triangle
cylinder	circle	congruent circle
square pyramid	square	smaller/larger similar square
cone	circle	smaller/larger circle

A non-parallel cut can produce a different shape. The orientation of the cutting plane is part of the description.

Build and annotate the model

Represent parallel cross-sections of three-dimensional objects and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Every slice matches the base:** Only slices parallel to the base have the stated relationship.
- **Prism and pyramid treated alike:** Prism sections stay congruent; pyramid sections scale.
- **Cross-section called a face of the original object:** It is created by the cut.
- **Orientation omitted:** State the cutting plane's direction.

Quick check

- Predict a prism slice.
- Compare prism/pyramid.
- Identify a cone section.
- Explain parallel.
- Describe a non-parallel cut.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.